Recall that the entangled state |*sing*> is defined as

$$|sing\rangle = \frac{1}{\sqrt{2}}\big(|ud\rangle - |du\rangle\big).$$

Let's look at the expectation values of $\sigma$ in this state. We have all the machinery we need to compute them. First, let's consider $\langle\sigma_z\rangle$:

$$\langle\sigma_z\rangle = \langle sing|\sigma_z|sing\rangle$$

$$= \quad \langle sing|\sigma_z \frac{1}{\sqrt{2}}\big(|ud\rangle - |du\rangle\big).$$

Here is where Eqs. 6.8 come in (along with Exercise 6.4, which completes this set of equations!). They tell us how $\sigma_z$ acts on each basis vector. The result is

$$\langle sing|\sigma_z|sing\rangle = \langle sing|\frac{1}{\sqrt{2}}\big(|ud\rangle + |du\rangle\big)$$

or

$$\langle\sigma_z\rangle = \frac{1}{2}\Big(\langle ud| - \langle du|\Big)\Big(|ud\rangle + |du\rangle\Big).$$

A quick inspection shows that this is equal to zero. Next, let's consider $\langle\sigma_x\rangle$: